Mantle Social Onchain Metrics AI Automation

Onchain Metrics Automation System

Technical Brief for Internal Developers & Engineers

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01: PURPOSE & PROBLEM STATEMENT

Background

The Mantle Social team currently monitors onchain metrics manually across six external data platforms: Dune Analytics, Artemis, DefiLlama, Growthepie, Nansen, and L2Beat. This process is time-consuming, inconsistent, and entirely dependent on individual team members remembering to check each platform at the right time.

As documented in the Mantle Social Sync, the Key Onchain Metrics & Market Signals pillar is rated Situational; strong when numbers are compelling but inconsistently surfaced because the data pull is manual and reactive. Social is filling the gap by self-sourcing, which is time that should not be Social's to spend.

The Problem

- No automated system exists to monitor key Mantle onchain metrics across platforms
- Social teams spend significant time manually checking six separate platforms
- Metrics are only posted when someone happens to notice a significant move
- No standardised threshold exists for what constitutes a post-worthy metric movement
- Design and copy teams cannot begin work until Social flags the data, creating downstream delays

- Coordination between Social, Design, and the internal data analyst is ad hoc and fragmented

The Goal

Build an automated onchain metrics monitoring system that:

1. Pulls live data from all six platforms on a defined cadence
2. Evaluates metric movements against pre-defined significance thresholds
3. Sends automated alert notifications to the Social and Design teams via Lark
4. Leverages Claude AI to generate a draft social copy suggestion alongside each alert
5. Reduces manual monitoring time to zero for the Social team

02: SYSTEM OVERVIEW

What We're Trying to Build

A fully automated system that monitors Mantle's onchain metrics, detects significant movements, and automatically pings the Social and Design teams in Lark with a notification and a suggested draft social post - with no manual checking required.

Once live, the expected human involvement is zero until a Lark alert arrives. At that point, the Social team reviews the alert, refines the draft copy if needed, and posts to @Mantle_Official. That's the only manual step in the entire process.

The Goal in One Sentence

Build it once. It runs forever. The Social team only gets involved when there's something genuinely worth posting and when they do, a draft post is already waiting.

High-Level Concept Flow

This is the intended end-to-end flow. Engineering team has full discretion on how best to implement each step - the below is the desired outcome, not a technical prescription:

Step	What Should Happen Automatically
Schedule fires	System wakes up on a defined schedule (e.g. every hour, every 6 hours, daily) — no human trig
Data is fetched	System pulls the latest metric values from all relevant platforms
Changes are calculated	System compares current values against historical snapshots to determine percentage movem all time windows
Claude evaluates everything	All metric data is sent to Claude in one call. Claude decides: is anything here post-worthy? If ye should we post, and what copy would work best?
Alert fires or doesn't	If Claude says post-worthy: a formatted Lark notification is sent automatically to the Social an team channels, including Claude's draft copy. If Claude says nothing significant: system logs it goes back to sleep.
Social team acts	Social team sees the Lark alert, reviews Claude's draft, refines if needed, approves, and posts @Mantle_Official. This is the only manual step.

Note for Engineering Team

The above is the desired outcome from a marketing and social operations perspective. We are not prescribing technical architecture, language, hosting environment, or tooling. The engineering team has full discretion to implement this in whatever way is most efficient, maintainable, and reliable.

The key requirements are:

1. Fully automated with no manual data feeding required
2. Claude AI is used as the signal evaluation and copy generation engine
3. Alerts are delivered via Lark
4. The system is reliable enough to catch ATH and milestone movements within one hour of them occurring

03: DATA SOURCES

Platform Integration Requirements

The system must pull data from the following six platforms. Each has different data availability, API access levels, and update frequencies. Engineering to confirm API availability and fallback scraping approach per platform.

Platform	URL	Access Method
Dune Analytics	dune.com/blockchains/mantle	Dune API (API key required) — public dashboard queries ava
Artemis	app.artemisanalytics.com/asset/mantle	Artemis API — confirm API key availability with team
DefiLlama	defillama.com/chain/mantle	Public REST API — no key required. Most reliable and recom primary source
Growthepie	growthepie.com/chains/mantle	Public API — confirm endpoint availability
Nansen	app.nansen.ai/macro/overview?chain=mantle	Nansen API — requires paid API access. Confirm team subsc
L2Beat	l2beat.com/scaling/projects/mantle	Public API / scraping — confirm structured data endpoints

Engineering Note — Primary vs. Fallback Sources

DefiLlama should be treated as the primary source for TVL and DeFi metrics given its public API, reliability, and industry-standard credibility. Other platforms should be used as supplementary or cross-verification sources. Where API access is unavailable or restricted, implement a structured web scraping fallback with appropriate rate limiting.

04: KEY METRICS TO MONITOR

Priority Metrics

The following metrics are the primary focus for monitoring and alerting. Note: no price-related metrics ($MNT token price) should be included in alerts. Price shilling is explicitly excluded from Mantle's social content policy.

Metric	Primary Source	Notes
TVL (Total Value Locked)	DefiLlama	Primary metric. Most frequently posted. Cross-verify with Nansen
Total Value Secured	L2Beat	Secondary metric. Post-worthy for major movements
Daily Active Users (DAU)	Artemis / Dune	Strong community signal. Use 7-day rolling average to smooth noise
Active Addresses	Dune / Artemis	Complement to DAU. Post-worthy for milestone round numbers
Stablecoin Supply	DefiLlama / Artemis	Key RWA thesis metric. Important for Mantle's distribution layer narrative
Stablecoin Market Cap	DefiLlama / Artemis	Closely related to Stablecoin Supply. Monitor together
Mantle Chain Transactions	Dune / Growthepie	Network activity signal. ATH or milestone round numbers are post-worthy
Stablecoin Transfer Volume	Dune / Artemis	Capital flow signal. Relevant for distribution layer narrative
DEX Volume	DefiLlama / Dune	Ecosystem activity. Post-worthy for significant weekly spikes
$MNT Volume	Artemis / Dune	Trading volume only — not price. Post-worthy for notable volume spikes
$MNT Market Cap	Artemis / CoinGecko	Market cap only — not price. Post-worthy for ATH or milestone movements

Excluded Metrics

The following must NOT be included in any automated alert or Claude-generated draft copy: $MNT token price, percentage price changes, price predictions or price comparisons. Mantle's social policy explicitly prohibits price shilling. Engineering must ensure these fields are filtered out of all data pulls and Claude prompts.

05: ALERT THRESHOLD LOGIC

When Should an Alert Fire?

Not every metric movement warrants a social post. The threshold evaluation engine must filter out minor fluctuations and only surface movements that are genuinely post-worthy. The following logic defines when an alert should be triggered.

5.1: Percentage Movement Thresholds

An alert fires when a metric moves by the following thresholds across the specified time windows:

Movement Threshold	Signal Strength	Recommended Action
~10% movement	Minor signal	Log internally. Do not alert unless combined with milestone (e.g. new ATH)
~15% movement	Moderate signal	Alert Social team. Optional post depending on context and market conditions
~20% movement	Strong signal	Alert Social and Design teams. High priority — aim to post within 24 hours
~30%+ movement	Major signal	Immediate alert. All-hands — Social, Design, and key advisor notification
New ATH (All-Time High)	Priority override	Alert fires regardless of percentage. Always post-worthy

Note: Feel free to adjust and add or refine based on the complexity; ideally, we would want more to be shown so the Social team can highlight and feature when we can.

5.2: Time Window Definitions

Each metric should be evaluated across the following time windows. Not all windows need to trigger alerts - prioritise 7D and MTD for frequency, longer windows for strategic narrative posts.

Time Window	Alert Priority
7 Days (7D)	HIGH — primary alert window. Most relevant for social posting cadence
Month to Date (MTD)	HIGH — good for monthly roundup content and narrative posts
1 Month (1M)	MEDIUM — useful for monthly comparison posts
3 Months (3M)	MEDIUM — useful for quarterly narrative and growth story posts
6 Months (6M)	LOW — strategic context only. Rarely post-worthy on its own
Year to Date (YTD)	MEDIUM — strong for end-of-quarter and milestone narrative posts
1 Year (1Y)	LOW-MEDIUM — annual milestone context
All Time	HIGH — ATH detection is always priority regardless of time window

5.3: Special Alert Conditions

In addition to percentage thresholds, the following conditions should trigger an automatic alert regardless of movement size:

- New All-Time High on any monitored metric
- Round number milestones (e.g. TVL crossing $500M, $750M, $1B, etc.)
- Metric drops significantly (>20% decline) — Social team needs to be aware even if not posting
- Two or more metrics hit significant thresholds simultaneously (double milestone moment)

06: ALERT NOTIFICATION FORMAT

Lark Notification Structure

Each alert sent to the Lark workspace must be structured and consistent so the Social and Design teams can immediately understand the signal and act on it. The following format is required for every alert.

Recommended Lark Alert Format

🟢 MANTLE METRICS ALERT (green for upwards movement & vice versa)

- 📊 Metric: TVL (Total Value Locked)
  - 📈 Movement: +66% (7D)
  - 💰 Current Value: $755M+
  - 🏆 Status: NEW ALL-TIME HIGH
  - 🛰️ Source: DefiLlama
  - ⏰ Detected: March 4, 2026 — 09:42 SGT
- ✍️ SUGGESTED DRAFT COPY (Claude AI):"New month. New ATH. Mantle's DeFi TVL surged to $755M+, recording a 66% growth in 7 days. When MoMNTum is inevitable, DeFi is inevitable, and we proved it."
- ⚡ ACTION REQUIRED:
- Social: Review draft copy and refine for posting
- Design: Prepare kinetic typography or metric card visual
- Target post window: Within 6 hours of alert

Lark Channel Structure

Alerts should be routed to the correct Lark channels to minimise noise and ensure the right people are notified:

Alert Type	Lark Channel
Strong signal (20%+ / ATH)	#social-metrics-alerts
Moderate signal (15-20%)	#social-metrics-alerts
Decline alert (>20% drop)	#social-metrics-alerts
Double milestone	#social-metrics-alerts

07: CLAUDE AI INTEGRATION (OPTIONAL)

Claude's Role in the Automated Pipeline

Claude is called automatically by the system every time the data ingestion layer completes a scheduled pull. It receives the raw metric data, evaluates it, and returns a structured JSON response that the alert delivery layer uses to decide whether to send a Lark notification and what it should contain. No engineer needs to write signal detection logic. No Social team member needs to check dashboards. Claude handles it all.

The Automated Flow: Step by Step

Step	What Happens	Who / What Does It
1. Scheduled trigger fires	Cron job or Lambda function triggers on schedule (hourly / 6-hourly / daily)	Engineering infrastructure — automatic
2. Data pull	System queries all six platforms via API or scraping and retrieves current metric values	Data ingestion layer — automatic
3. Snapshot comparison	System retrieves previous snapshots from database and calculates percentage changes across all time windows	Data ingestion layer — automatic
4. Claude API call	System sends raw data package to Claude API with full context: current values, percentage changes, time windows, historical snapshots, ATH flags	System calls Claude API — automatic
5. Claude evaluates	Claude assesses signal strength, contextual significance, multi-metric correlations, and anomalies. Decides: post-worthy or not.	Claude AI — fully automatic
6. Claude generates response	If post-worthy: Claude returns structured JSON with alert metadata, signal rating, recommended action, and draft copy. If not: Claude returns a dismiss signal.	Claude AI — fully automatic
7. Alert routing	Alert delivery layer reads Claude's JSON response. If post-worthy: formats and sends Lark card to correct channel. If dismissed: logs silently.	Alert delivery layer — automatic
8. Human approval	Social team receives Lark alert, reviews Claude's draft copy, refines if needed, approves, and posts to @Mantle_Official	Social team — intentionally manual

Claude API Configuration

- Model: claude-sonnet-4-6
- Max tokens: 800 (enough for structured JSON response + draft copy)
- Temperature: 0.7 (balanced between consistency and creativity)
- Response format: Structured JSON — see Section 7.2
- System prompt: Pre-loaded with Mantle's brand voice, signal evaluation criteria, and content rules — see Section 7.3

7.1: Claude Input Data Package

Every Claude API call must include the following data package, so Claude has full context to make accurate signal evaluations:

Claude Input Data Package (Engineering Reference){ "timestamp": "2026-03-04T09:42:00+08:00", "metrics": [{ "name": "TVL", "current_value": 755000000, "formatted_value": "$755M", "source": "DefiLlama", "changes": { "7d": "+66.2%", "mtd": "+71.4%", " "ytd": "+89.3%", "all_time_high": true, "previous_ath": 720000000 } }, { "name": "Aave Total Market Size", "current_value": 10500 "formatted_value": "$1.05B", "source": "Aave", "changes": { "7d": "+31.2%", "mtd": "+40.1%", "all_time_high": true } }], "market_ "Multiple metrics hitting ATH simultaneously"}

7.2: Claude Structured JSON Output

Claude must return a structured JSON response that the alert delivery layer can parse and act on automatically. The system prompt must instruct Claude to always respond in this exact format:

Claude Output JSON Format (Engineering Reference){ "post_worthy": true, "signal_strength": "STRONG", "signal_rating": 9, "alert_priori "reason": "TVL hitting new ATH at $755M+ with 66% 7D growth. Simultaneous Aave ATH creates double milestone moment — higher impact in one post.", "recommended_action": "Combine both metrics into single post for maximum impact. Alert Social and Design immediately.", "combine_metrics": ["TVL", "Aave Total Market Size"], "suppress_individual_alerts": true, "draft_copy": { "tweet": "New month. New ATH. Ac that two.\n\nMantle crossed $1B in total lending and borrowing on @aave.\nMantle's DeFi TVL surged to $755M+, recording a 66% growth in \n\nWhen MoMNTum is inevitable, DeFi is inevitable — and we proved it. 👀", "character_count": 276, "visual_suggestion": "Clean dual ATH Aave + $755M TVL side by side. Kinetic typography format." }, "sources": ["DefiLlama", "Aave"], "cooldown_recommended": "48 hours before alert"}

7.3: Claude System Prompt - Full Specification

The following system prompt must be hardcoded into every Claude API call. This defines Claude's role, evaluation criteria, output format, and brand voice rules:

Claude System Prompt (Engineering — Copy Exactly) You are the automated signal detection and social copywriting engine for Mantle's @Mantle_Official X account. You receive onchain metric data from six platforms and make two decisions: (1) is this signal post-worthy, and (2 should the social post say.

SIGNAL EVALUATION RULES:— A signal is post-worthy if it meets one or more of these criteria: 15%+ movement in 7D window; 20%+ movem window; new all-time high on any metric; round number milestone (e.g. $500M, $750M, $1B); two or more metrics hitting significant thresho simultaneously.— A signal is NOT post-worthy if: it is a recovery bounce after a recent decline; it reverses a signal already posted about withi movement is within normal daily variance; the metric is price-related ($MNT token price changes are always excluded).— When multiple met moving simultaneously, always recommend combining them into a single post rather than separate alerts.— Always flag when a cooldown p recommended before the next alert for the same metric.

BRAND VOICE RULES FOR DRAFT COPY:— Write in the style of Coinbase's X account: confident, clean, minimal, dry wit, no corporate speak.— mention $MNT token price or percentage price changes under any circumstances.— Never use: 'thrilled to announce', 'excited to share', 'LFG' 'WAGMI', 'up only'.— MoMNTum must always be capitalised exactly as: MoMNTum.— Maximum one emoji per post. Prefer 👀 or 📈 for metric Always tag data sources (e.g. per @DefiLlama).— Posts must be under 280 characters.— Write in sentence case. Short declarative sentences. does the work — not adjectives.

OUTPUT FORMAT:— Always respond in valid JSON matching the exact schema provided. Never include text outside the JSON object.— If not set post_worthy to false and omit draft_copy entirely.— signal_rating is 1-10 where 10 is the most post-worthy signal possible (e.g. first-ever milestone).

08: DATA PULL CADENCE & SCHEDULING

How Often Should Data Be Pulled?

Different metrics update at different frequencies. The system should be configured to pull data at the appropriate cadence per metric type to avoid unnecessary API calls while ensuring timely alert detection.

Cadence	Metrics	Rationale
Every 1 hour	TVL, Stablecoin Supply, DEX Volume	Fast-moving metrics. ATH detection requires nea time monitoring
Every 6 hours	Daily Active Users, Active Addresses, Chain Transactions	Daily metrics — 6-hour cadence balances timelir with API rate limits
Every 24 hours	Total Value Secured, Stablecoin Transfer Volume, $MNT Volume, $MNT Market Cap	Slower-moving metrics. Daily snapshot is sufficie
Weekly (Monday 09:00 SGT)	All metrics — 7D window evaluation	Weekly summary evaluation for MTD and 7D trer alerts
Monthly (1st of month, 09:00 SGT)	All metrics — MTD, 1M, 3M, YTD evaluation	Monthly summary for roundup content and long term trend alerts

Engineering Note — Rate Limiting

Be mindful of API rate limits per platform. DefiLlama public API is generous but should still be cached where possible. Dune API has stricter rate limits on free tiers — confirm the team's subscription level. Implement exponential backoff on all API calls. Cache responses for at least 30 minutes to avoid redundant calls.

09: LARK INTEGRATION SPECIFICATION

Why Lark?

Lark (Feishu) is Mantle's current primary internal communication platform. The alert system should integrate natively with Lark using Lark's Webhook and Bot API to deliver formatted alert cards directly into designated Lark channels. No new tools or platforms are required.

Integration Method

- Primary: Lark Incoming Webhook — simplest implementation for formatted card alerts
- Secondary: Lark Bot API — for more interactive alerts with action buttons (e.g. 'Mark as Posted', 'Dismiss')
- Recommended implementation: Start with Webhook for v1, upgrade to Bot API for v2 once core system is stable

Lark Card Format Requirements

Each Lark alert card must include the following fields in a structured, readable format:

- Alert severity indicator (colour-coded: 🟢 Strong / 🟡 Moderate / 🔴 Decline)

- Metric name and current value
- Percentage movement and time window
- ATH or milestone status flag if applicable
- Source platform with link
- Timestamp in SGT (Singapore Time — Mantle's primary operating timezone)
- Claude AI draft copy block
- Action items for Social and Design teams

10: IMPLEMENTATION PHASES

Recommended Build Sequence

To minimise risk and allow for iterative testing, the system should be built in three phases:

Phase	Scope	Success Criteria
Phase 1 — Foundation (Weeks 1-2)	DefiLlama API integration only. TVL monitoring. Basic percentage threshold logic. Lark Webhook alerts (no Claude yet). Manual testing with Social team.	Social team receives accurate TVL alerts in Lark within 1 hour of a 15%+ movement. Zero false positives in first week of testing.
Phase 2 — Expansion (Weeks 3-4)	Add remaining five platforms. Add all 11 metrics. Add Claude AI draft copy generation. Add ATH and milestone detection. Refine alert formatting based on Phase 1 feedback.	All 11 metrics monitored across all 6 platforms. Claude draft copy included in every alert. Social team confirms drafts are useful and brand-aligned.
Phase 3 — Optimisation (Weeks 5-6)	Add multi-metric correlation alerts (double milestone detection). Add Lark Bot API for interactive alert actions. Add weekly and monthly summary digests. Performance tuning and rate limit optimisation.	System runs fully autonomously with zero manual intervention from Social team. Alert-to-post time reduced to under 4 hours for strong signals.

11: OPEN QUESTIONS FOR ENGINEERING

Items Requiring Engineering Confirmation

The following items need to be confirmed or decided by the engineering team before build begins:

1. API access levels — confirm which platforms have active API keys or subscriptions (particularly Nansen and Dune)
2. Lark Bot permissions — confirm the team has admin access to create and configure Lark Bots and Webhooks in the relevant workspace

3. Claude API access — confirm the team has an active Anthropic API key with sufficient token budget for the expected call volume
4. Hosting environment — where will this system be hosted? (AWS Lambda / internal server / other)
5. Data storage — historical metric data needs to be stored to enable percentage change calculations. Confirm database preference (PostgreSQL / DynamoDB / other)
6. Scraping fallback — for platforms without reliable APIs (Growthepie, L2Beat), confirm whether scraping is acceptable and legally permissible under their ToS
7. Alert suppression logic — if a metric has already been posted about recently, should the next alert for the same metric be suppressed for a cooldown period? If yes, what is the cooldown window?
8. On-call coverage — who is the engineering point of contact if the system goes down or generates false positive alerts outside of business hours?

12: SUCCESS METRICS

How We Measure Success

The system will be considered successful when the following outcomes are consistently achieved:

1. Data Ingestion Layer	Scheduled jobs pull raw metric data from all six platforms at defined intervals and store snapshots in a database	Engineers — build once, runs automatically forever
2. Claude AI Intelligence Layer	Receives raw data, evaluates signal strength, detects anomalies and ATHs, correlates multi-metric movements, decides post-worthiness, generates draft copy, and returns a structured response	Claude API — fully automated, no engineering logic required
3. Alert Delivery Layer	Receives Claude's structured response and routes post-worthy alerts to the correct Lark channel in the correct format	Engineers — build once, runs automatically forever
4. Human Approval Step	Social team receives Lark alert with Claude's draft copy, reviews, refines, and approves before posting to @Mantle_Official	Social team — intentionally manual. This is the only human touchpoint in the entire flow
False positive rate	Under 5% — alerts that fire but are deemed not post-worthy by Social team	
Claude draft quality	Social team uses draft as base for final copy in >70% of alerts (requires minimal rewriting)	
System uptime	99%+ — alerts must be reliable. Missed ATH alerts are unacceptable	
Coverage	100% of 11 monitored metrics covered across all 6 platforms within Phase 2 completion	

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For questions on this document, contact Wayne or mk.chin